

Cloud watch

Amazon CloudWatch is a monitoring and observability service in AWS that helps you track, analyze, and respond to your AWS resources and applications in real time.

👁️👂 Simple Definition

CloudWatch = “Eyes and Ears of AWS” 👁️👂👁️👂👁️👂👁️👂

It continuously monitors your resources and alerts you when something goes wrong.

Key Features

📊📊📊 Metrics Monitoring

- Tracks performance data like:
 - CPU usage (EC2)
 - Memory (via custom metrics)
 - Disk I/O
 - Example: Monitor CPU utilization of your EC2 instance
-

📖📖📖 Logs Management

- Collect and store logs from:
 - EC2
 - Lambda
 - Applications
 - Helps in debugging and troubleshooting
-

🚨🚨🚨 Alarms

- ☐ Set thresholds and get alerts ☐ Example:
 - If CPU > 80% → send notification (SNS)
 - Auto-scale EC2 instances

📊 Dashboards

- Create visual dashboards
- Combine multiple metrics in one view

5. ☐ Events (EventBridge integration) ☐ React to changes automatically ☐

Example:

- o Trigger Lambda when an EC2 stops

🕒 Real-Time Example (DevOps)

Imagine you deployed a web app on EC2:

- CloudWatch monitors CPU usage 📊
- CPU goes above 80% 🚨
- Alarm triggers → Auto Scaling launches new instance ☐
- Your app stays stable without downtime

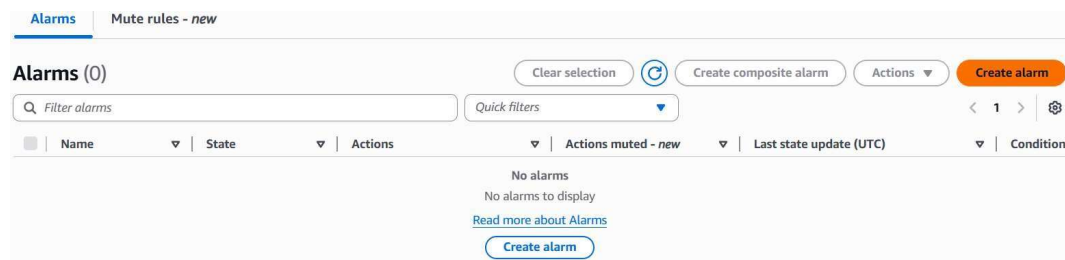
🔗 Common AWS Services Integrated

- EC2
- S3
- Lambda
- RDS
- Auto Scaling

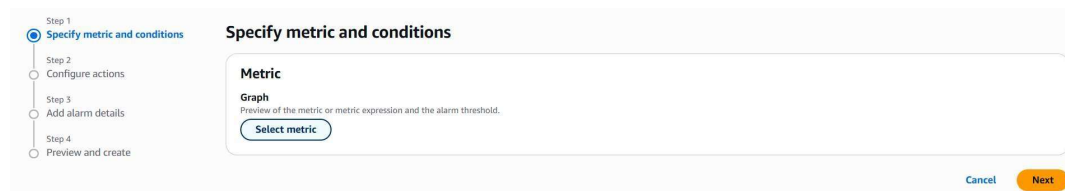
🐛 Why CloudWatch is Important

- Improves system reliability
- Helps in troubleshooting
- Enables automation
- Provides real-time insights

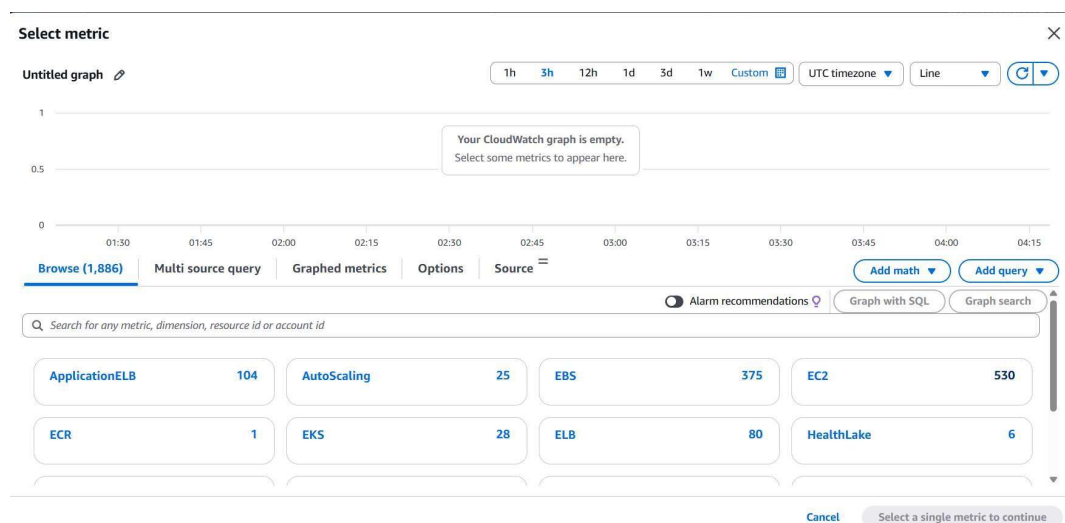
Step 1 : create the alarm



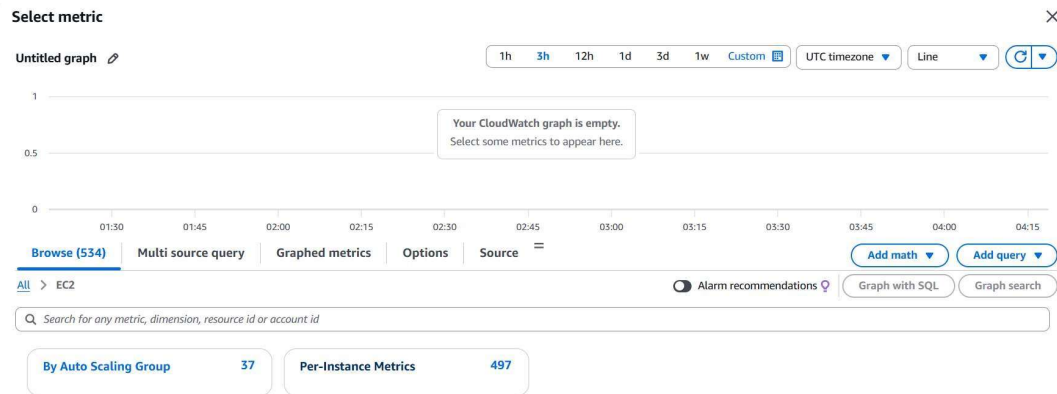
Step 2 : select the metric



Step 3 : select the EC2 metric



Step 4 : select the per-instance metrics



Step 5 : select our instance

Search for any metric, dimension, resource id or account id

i-Oedc2205137052e4c Clear filters

☐	Instance name 18/18	Instanceid	Metric name	Alarms
☐	linux	i-Oedc220513705...	CPUUsage	No alarms
☐	linux	i-Oedc220513705...	CPUCreditUsage	No alarms
☐	linux	i-Oedc220513705...	MetadataNoToken	No alarms
☐	linux	i-Oedc220513705...	CPUCreditBalance	No alarms

Cancel Select a single metric to continue

Step 6 : select the CPU metric

Select metric

CPUUtilization

1h 3h 12h 1d 3d 1w Custom UTC timezone Line

No data available.
Try adjusting the dashboard time range.

0 0.5 1

10:45 11:00 11:15 11:30 11:45 12:00 12:15 12:30 12:45 13:00 13:15 13:30 13:45

CPUUtilization

Browse (18) Multi source query Graphed metrics (1) Options Source

Add math Add query

☐	linux	i-Oedc220513705...	EBSReadBytes	No alarms
☐	linux	i-Oedc220513705...	EBSByteBalance%	No alarms
☐	linux	i-Oedc220513705...	EBSWriteBytes	No alarms
☒	linux	i-Oedc220513705...	CPUUtilization	No alarms
☐	linux	i-Oedc220513705...	EBSReadOps	No alarms
☐	linux	i-Oedc220513705...	EBSWriteOps	No alarms
☐	linux	i-Oedc220513705...	InstanceEBSThroughputExceededCheck	No alarms
☐	linux	i-Oedc220513705...	InstanceEBSIOPSExceededCheck	No alarms
☐	linux	i-Oedc220513705...	NetworkOut	No alarms

Cancel Select metric

Step 7 : create the sns topic

Amazon SNS > Topics > Create topic

Create topic

Details

Type [Info](#)
Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name
mytopic
Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional [Info](#)
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.
My Topic
Maximum 100 characters.

► **Encryption - optional** [Info](#)
Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

► **Access policy - optional** [Info](#)
This policy defines who can access your topic. By default, only the topic owner can publish or subscribe to the topic.

Step 8 : create the subscrip on

Amazon SNS > Topics > mytopic

Amazon SNS

- Dashboard
- Topics
- Subscriptions
- ▼ Mobile
 - Push notifications
 - Text messaging (SMS)

Topic mytopic created successfully.
You can create subscriptions and send messages to them from this topic. [Publish message](#)

Notifications 0 0 1 2 0

mytopic [Edit](#) [Delete](#) [Publish message](#)

Details

Name mytopic	ARN arn:aws:sns:ap-south-1:089665204192:mytopic	Display name -	Type Standard
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Topic owner
089665204192

Subscriptions (0) [Edit](#) [Delete](#) [Request confirmation](#) [Confirm subscription](#) [Create subscription](#)

Search

ID	Endpoint	Status	Protocol
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Step 9 : add your email

Amazon SNS > Subscriptions > Create subscription

Create subscription

Details

Topic ARN
arn:aws:sns:us-east-1:051741041276:mytopic

Protocol
The type of endpoint to subscribe
Email

Endpoint
An email address that can receive notifications from Amazon SNS.
hharan1899@gmail.com

After your subscription is created, you must confirm it. [Info](#)

► **Subscription filter policy - optional** [Info](#)
This policy filters the messages that a subscriber receives.

► **Redrive policy (dead-letter queue) - optional** [Info](#)
Send undeliverable messages to a dead-letter queue.

[Cancel](#) [Create subscription](#)

Step 10 : accept the subscrip on in the email

The screenshot shows the Amazon SNS console interface. On the left, the navigation menu includes 'Amazon SNS', 'Dashboard', 'Topics', 'Subscriptions', and 'Mobile'. The main content area displays a green notification banner at the top stating 'Subscription to mytopic created successfully. The ARN of the subscription is arn:aws:sns:us-east-1:051741041276:mytopic:0f7985ce-4520-4e73-8310-3fd92aea6f8f.' Below this, the subscription details are shown for 'Subscription: 0f7985ce-4520-4e73-8310-3fd92aea6f8f'. The details include the ARN, endpoint (hharan1899@gmail.com), topic (mytopic), and subscription principal. The status is 'Pending confirmation' and the protocol is 'EMAIL'. Below the details, there are tabs for 'Subscription filter policy' and 'Redrive policy (dead-letter queue)'. The 'Subscription filter policy' tab is active, showing a message that no filter policy is configured for this subscription.

Amazon SNS > Topics > mytopic > Subscription: 0f7985ce-4520-4e73-8310-3fd92aea6f8f

Subscription to mytopic created successfully.
The ARN of the subscription is arn:aws:sns:us-east-1:051741041276:mytopic:0f7985ce-4520-4e73-8310-3fd92aea6f8f.

Subscription: 0f7985ce-4520-4e73-8310-3fd92aea6f8f [Edit] [Delete]

Details

ARN
arn:aws:sns:us-east-1:051741041276:mytopic:0f7985ce-4520-4e73-8310-3fd92aea6f8f

Endpoint
hharan1899@gmail.com

Topic
[mytopic](#)

Subscription Principal
arn:aws:iam::051741041276:root

Status
Pending confirmation

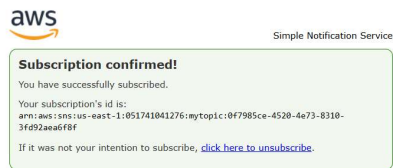
Protocol
EMAIL

Subscription filter policy | Redrive policy (dead-letter queue)

Subscription filter policy [Info](#)
This policy filters the messages that a subscriber receives.

No filter policy configured for this subscription.
To apply a filter policy, edit this subscription.

[Edit]



Step 11 : The subscrip on is confirmed

The screenshot shows the Amazon SNS console with the 'Subscriptions' tab selected. It displays a table with one subscription listed, which is confirmed. The subscription ID is 0f7985ce-4520-4e73-8310-3fd92aea6f8f, the endpoint is hharan1899@gmail.com, and the status is 'Confirmed'. The protocol is 'EMAIL' and the topic is 'mytopic'. The console also shows buttons for 'Edit', 'Delete', 'Request confirmation', 'Confirm subscription', and 'Create subscription'.

Amazon SNS > Subscriptions

Subscriptions (1) [Edit] [Delete] [Request confirmation] [Confirm subscription] [Create subscription]

Q Search

ID	Endpoint	Status	Protocol	Topic
0f7985ce-4520-4e73-8310-...	hharan1899@gmail.com	Confirmed	EMAIL	mytopic

Step 12 : select the SNS topic

CloudWatch > Alarms > Create alarm

Step 2: **Configure actions**

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

mytopic

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)
hari636932@gmail.com - View in SNS Console

Add notification

Step 13 : add alarm details

CloudWatch > Alarms > Create alarm

Specify metric and conditions

Step 2: Configure actions

Step 3: **Add alarm details**

Step 4: Preview and create

Add alarm details

Name and description

Alarm name
danger for high cpu

Alarm description - optional [View formatting guidelines](#)

Edit **Preview**

Open the AWS Systems Manager console.
In the Regions list, select your AWS Region.
In the navigation pane, expand Shared Resources, and then choose Documents.
In the search bar, enter [AWSsupport-TroubleshootWindowsPerformance](#), and then select the runbook.

Up to 1024 characters (282/1024).

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Tags - optional [info](#)

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel **Previous** **Next**

Step 14 : the alarm was created

CloudWatch > Alarms

CloudWatch

Successfully created alarm danger for high cpu. [View alarm](#)

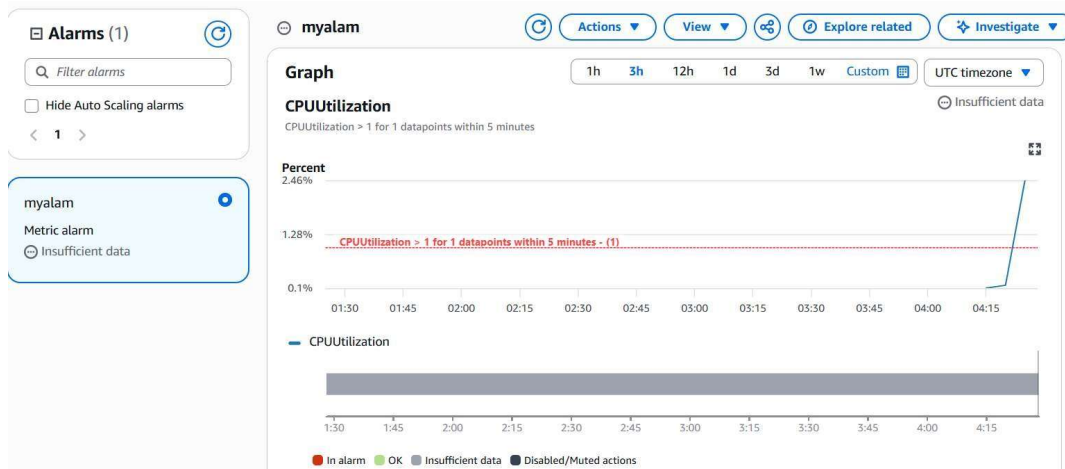
Alarms Mute rules - new

Alarms (1) [Clear selection](#) [Create composite alarm](#) [Actions](#) [Create alarm](#)

[Filter alarms](#) [Quick filters](#)

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Condition
☐	danger for high cpu	☐ Insufficient data	☒ Actions enabled	-	2026-04-01 14:05:24	CPUUtiliza minutes

Step 15 : the alarm is triggered



Step 16 : the alarm state is in the state of “in alarm”

